Coagulation and Flocculation: Hydrolyzing Metal Salts

Objectives

To conduct jar tests on a natural surface water in order to estimate an optimum dosage of aluminum sulfate for the removal of suspended matter or color. To observe the rate of floc formation and sedimentation.

Apparatus

1. Six-place laboratory stirrer (jar test apparatus)
2. Turbidimeter
3. pH Meter
4. Assorted measurement pipettes (1, 5, 10 ml) and volumetric dispenser to 50 ml
5. Plastic cups for coagulant additions, 10 ml.

Materials

1. Aluminum sulfate solution, 10 mg $Al_2(SO_4)_3 \cdot 14H_2O$/ml, 1 liter
2. Sodium carbonate[1] 10 g/L as $CaCO_{3\ eq}$.
3. Bentonite suspension, 1000 mg/l, made from deionized, 10.8 L.
4. Standard buffer solutions.

Procedure

Part 1: Determination of Optimum Coagulant Dosage

1. Analyze Bentonite Suspension for pH and turbidity. Record both the water temperature at time of test and ambient air temperature.
    - Turbidity: Fill the sample cell to the line with approximately 15-mL of sample. Cap the sample cell. Clean the outside of the cell with a soft, lint-free cloth to remove fingerprints and water spots (if needed apply a thin film of silicone oil and wipe with a soft, lint-free cloth to obtain an even film over the entire surface.) Press I/O button. Put the sample cell in the instrument cell compartment so the white diamond or assigned orientation mark aligns with the raised mark in front of the cell compartment. Close the cover. Select manual or automatic range by pressing the RANGE key. The display will show "AUTO RNG" when the instrument is in the automatic range selection. Select signal averaging mode by pressing the SIGNAL AVERAGE key. The display will show "SIG AVG" when the instrument is using averaging. Use signal average mode if the sample causes a noisy signal (display changes constantly). Press READ. The display will show "- - - - NTU" then the turbidity value in NTU. Record the turbidity after the lamp symbol in the lower left corner of the screen turns off.
    - pH: immerse probe and temperature sensor in solution. Press read on pH meter. Wait for reading to stabilize and record the pH.

2. Place 600 milliliter aliquots in 1-liter beakers on the six-jar laboratory stirrer and check stirrer operation (200 rpm). A light table will facilitate viewing of the contents of the beakers.

3. The initial alkalinity of the water is 0 mg/L as $CaCO_3$ equivalent. Augment the alkalinity of the water by adding 30 mL of Na_2CO_3 using a graduated cylinder to transfer the amount of Na_2CO_3 to each 1 L beaker with sample. Measure and record the pH of each Jar Tester.

4. Prepare the coagulant dosage using the 50 mL conical containers and 50 mL graduated cylinder. In five different conical containers, measure out 4, 8, 12, 16 and 20 mL of aluminum or ferric sulfate solution. **You will add (in the next step) the coagulant all at once and after alkalinity adjustment.**

5. After alkalinity has been added and the speed has been increased to 200 rpm, now *(quickly)* add 4 to 25 mL of

[1] 3.0 g Na_2CO_3/500mL

aluminum or ferric sulfate solution to the five beakers keeping one beaker as a control and start a 15 seconds rapid mix at 200 rpm. This step should be done as quickly as possible. Be sure you are ready to add the coagulant and start rapid mix.

6. Flocculate at 60 rpm speed for 20 minutes to represent design conditions. Record the elapsed time before a visible floc is formed. If large flocs are formed, it may be desirable to reduce the paddle speed. Note the size and appearance of the floc formed.

7. After flocculation, remove the paddles and settle for 15 minutes.

8. Measure the turbidity of the liquid in each jar by sampling at the supernatant (at 2¼" or 6 cm below the water surface), taking care not to disturb the sediment in sampling. Measure the pH of the supernatant and the depth of sludge (use ruler supplied). Estimate the volume of sludge which might be produced by treating 1 million gallons of the test water.

9. Based on these results, select the optimum dosage based on supernatant clarity (both low turbidity) and settleability of floc.

10. If satisfactory results are not obtained, repeat the jar test experiment with another range of coagulant doses. The test also may be repeated, if time allows, to further narrow the range of dosages near the optimum. Estimate the range of dosages over which this coagulant would be effective.

Part 2: Effects of Settling Time – Type II flocculated particle settling.

1. Type II (or flocculent) settling is sedimentation of suspensions that flocculated particles. All six beakers will have the same chemical additives.

2. No pH adjustment is needed for this experiment. All beakers should be amended with the required alkalinity before the addition of the optimum coagulant dose. Analyze Bentonite Suspension in each jar for turbidity, pH and temperature. Record both the water temperature and pH at time of test.

3. In six different containers, measure out the optimum dosage (determined from previous Laboratory 1) of aluminum or ferric sulfate solution. Prepare identical (optimum) coagulant dosages for all six (6) beakers. Now add the coagulant to each of the six (6) beakers and start a one-minute rapid mix at 200 rpm. This step should be done as quickly as possible. Be sure you are ready to add the coagulant and start rapid mix.

4. After one-minute rapid mix, reduce speed to 40 rpm. Allow flocculation for 10 minutes. At the end of the flocculation time, stop stirring and begin settling cycle. In order to better approximate the clarification, it is good practice to allow a lag time of about 30 seconds from the time of stopping the flocculation to the beginning of settling. This is about the time required for the fast rotary motion to subside. In a plant, it is about the time required for the flocculated water to go from the flocculation to settling basins before settling begins.

5. Allow for 5, 10, 15, 20, 25, 30 minutes for settling. For each settling time, measure turbidity by pulling a sample from 2¼" or 6 cm below the water surface. Repeat until the end of the total settling time.

6. Measure the depth of sludge in the beaker.

Report – submit a final report that includes a discussion of the following analysis and address the problems and questions.

1. Prepare tables which facilitate comparisons of coagulant dosages with pH, turbidity and sludge production. Plot turbidity versus coagulant dosage as part of the analysis. Plot turbidity at different flocculation times and another plot with different settling times to determine the influence of floc formation and settling characteristics on the selection of coagulant dosage.

2. Develop a settling plot using appropriate average turbidity and standard deviations.